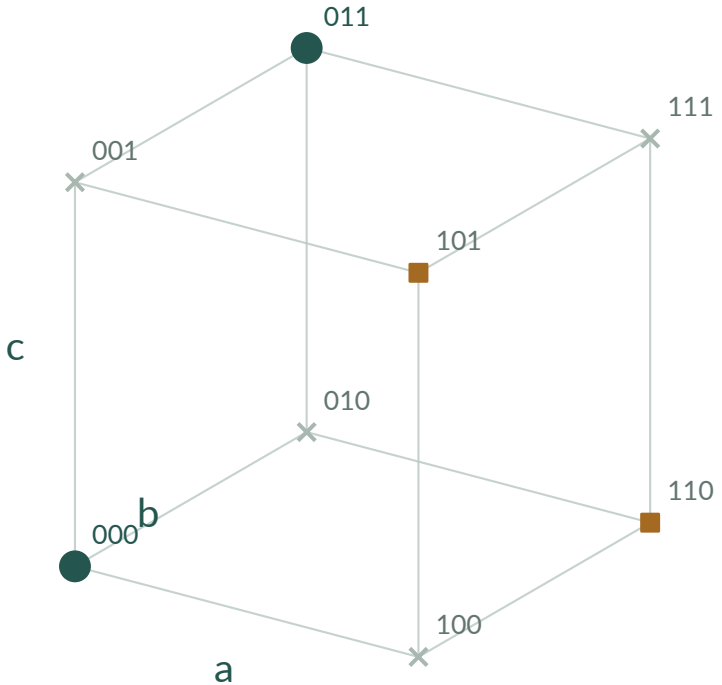
$$(a,b,c)\in\mathbb{F}_2^3,\quad a+b+c=0,\quad a=0$$



8 states

4 satisfy parity

2 preserve a = 0